Анализ теории измерений Раша в R

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Анализ теории измерений Раша в R: полное практическое руководство по надежным психометрическим измерениям 📊📚

Введение 🚀

Современная инженерия, здравоохранение, психология, образование и контроль качества на производстве — все эти сферы опираются на **точные измерения**. Исследователям нужны объективные, надежные и научно обоснованные системы измерения, будь то оценка компетентности сотрудников, измерение удовлетворенности клиентов, оценка инженерных навыков или проверка образовательных экзаменов.

Традиционные методы подсчета баллов предполагают, что каждый вопрос в равной степени влияет на итоговый результат. К сожалению, это предположение редко соответствует действительности. Некоторые вопросы по своей природе проще, в то время как другие гораздо сложнее. Кроме того, у респондентов разный уровень подготовки.

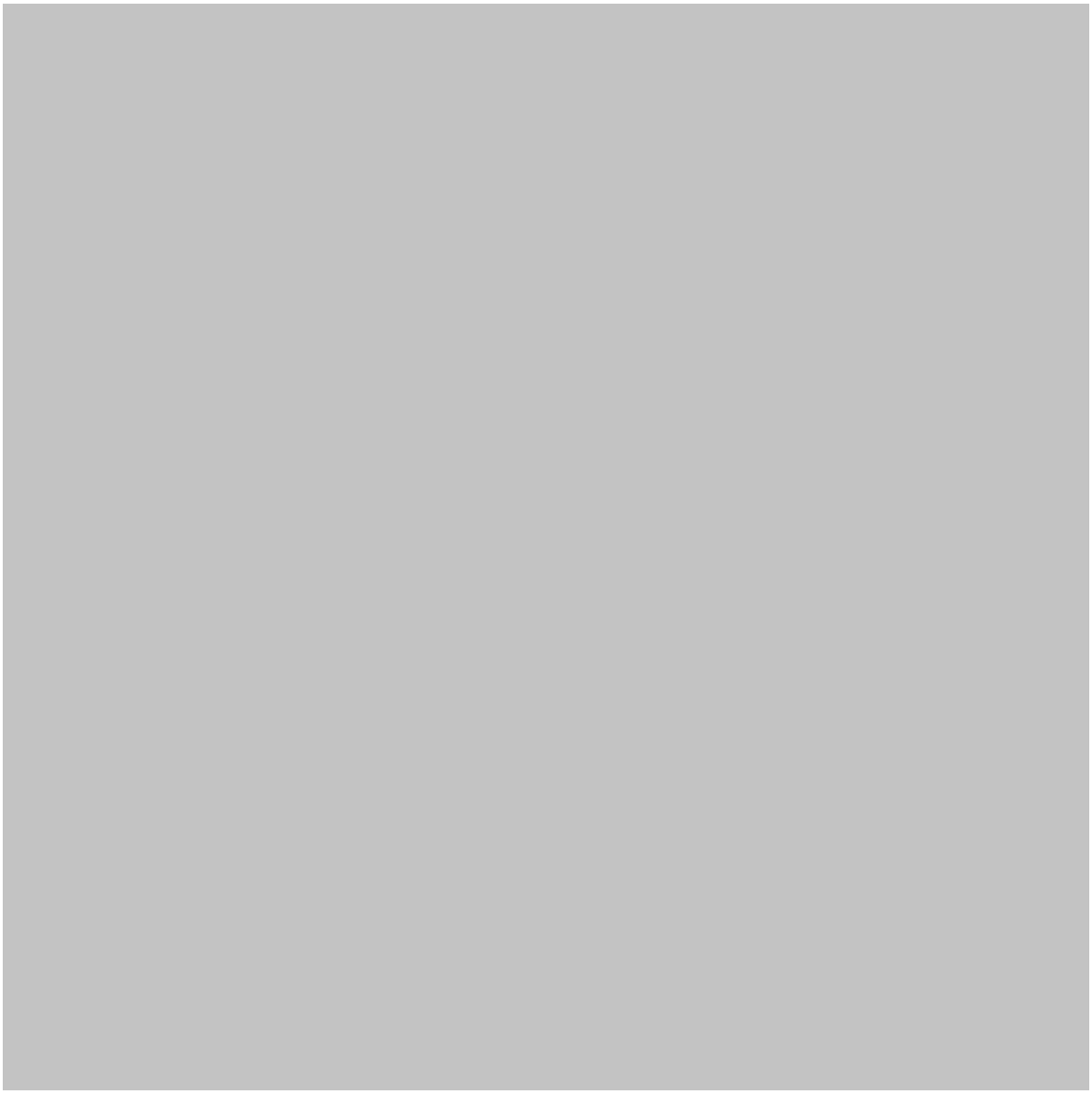
Теория Раша (RMT) решает эту проблему, преобразуя исходные порядковые оценки в измерения интервального уровня, что позволяет проводить более содержательный статистический анализ.

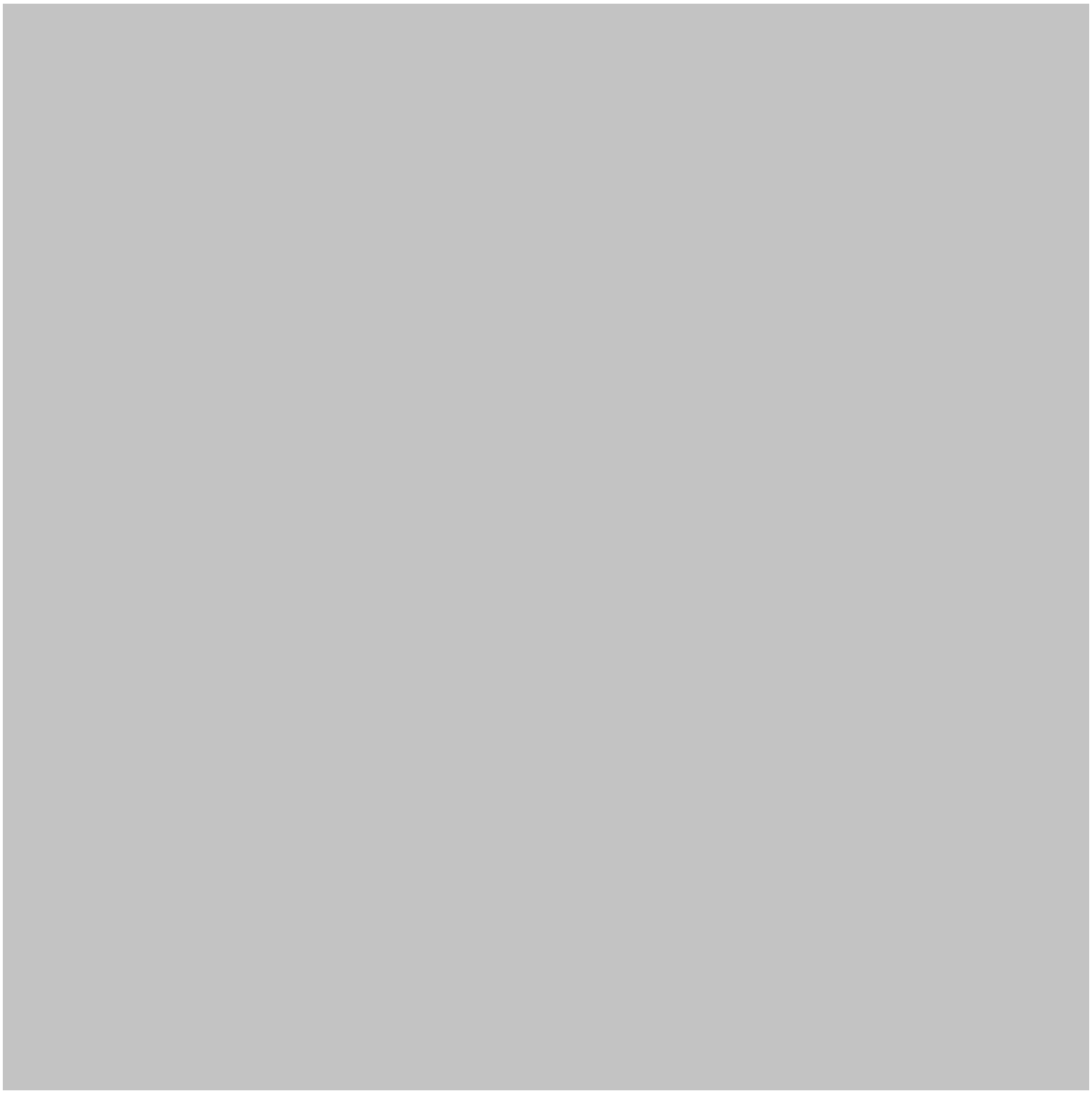
Сегодня **R** стал одним из самых популярных в мире языков статистического программирования, поскольку он предоставляет мощные пакеты с открытым исходным кодом для анализа Раша, в том числе:

- eRm
- TAM
- mirt
- psych
- WrightMap

В этом подробном руководстве теория Раша излагается от начального до продвинутого уровня. В нем показано, как инженеры, исследователи, статистики и специалисты по анализу данных могут проводить анализ Раша в R.







Теория предыстории📖

Модель Раша была предложена датским математиком **Георгом Рашем** в 1960–х годах.

Ее цель была проста, но революционна:

> *Оценка должна зависеть от взаимодействия между **способностями человека** и **сложностью задания**, а не только от первичных баллов.*

В отличие от классической теории тестов (CTT), анализ Раша рассматривает измерения аналогично физическим наукам.

Например:

- Термометр измеряет температуру.
- Линейка измеряет длину.
- Модель Раша измеряет скрытые характеристики, такие как:
    - Knowledge
    - Competency
    - Attitude
    - Engineering skills
    - Safety awareness
    - Customer satisfaction

The model converts ordinal questionnaire responses into interval-scale estimates.

Definition 🎯

Rasch Measurement Theory (RMT) is a probabilistic measurement model that estimates:

- Person ability (θ)
- Item difficulty (β)

using logistic probability equations.

Instead of assuming all questions have equal importance, Rasch estimates each item's difficulty independently.

As a result:

✅ Fair comparisons become possible.

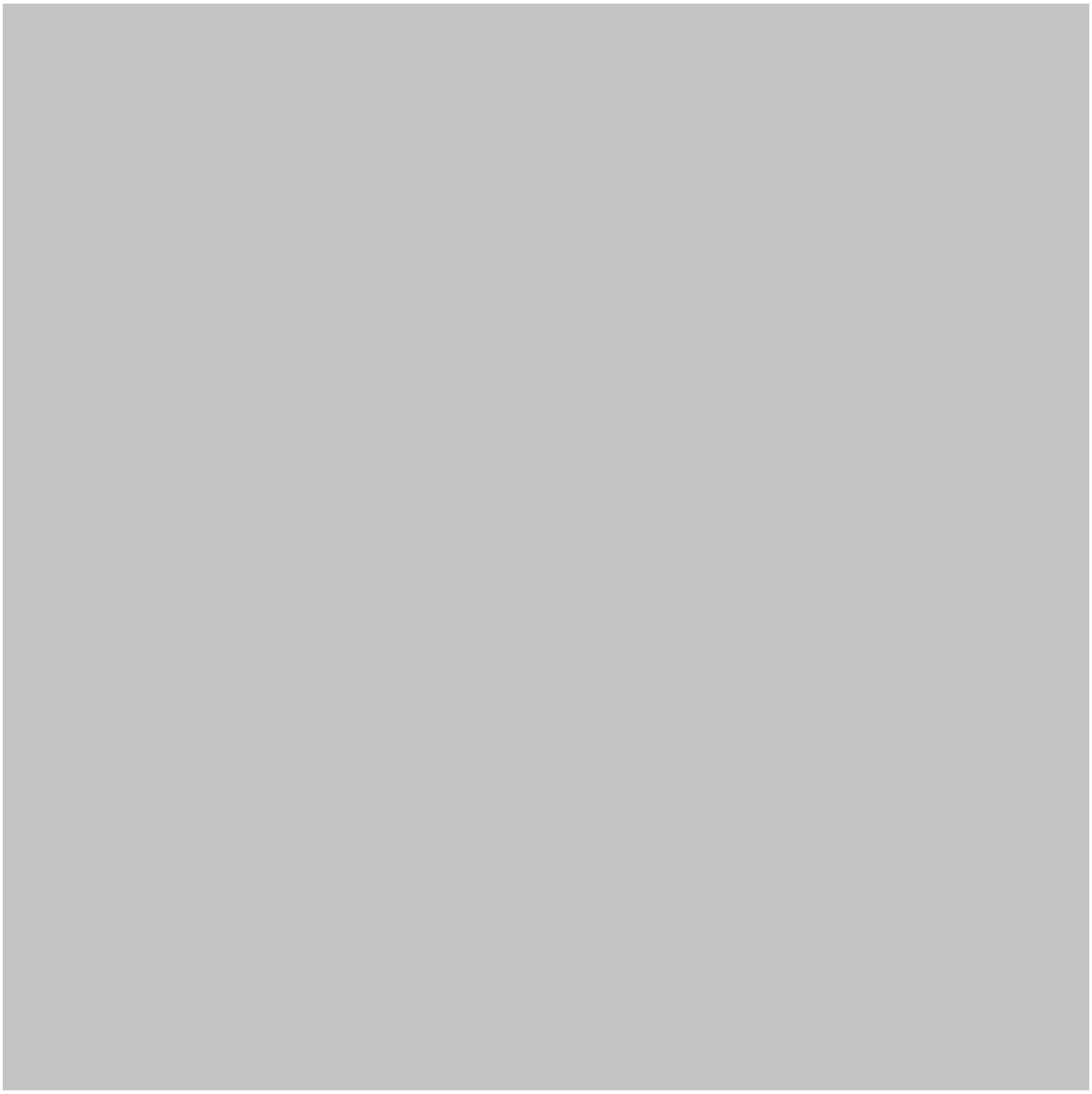
📊 Missing data become less problematic.

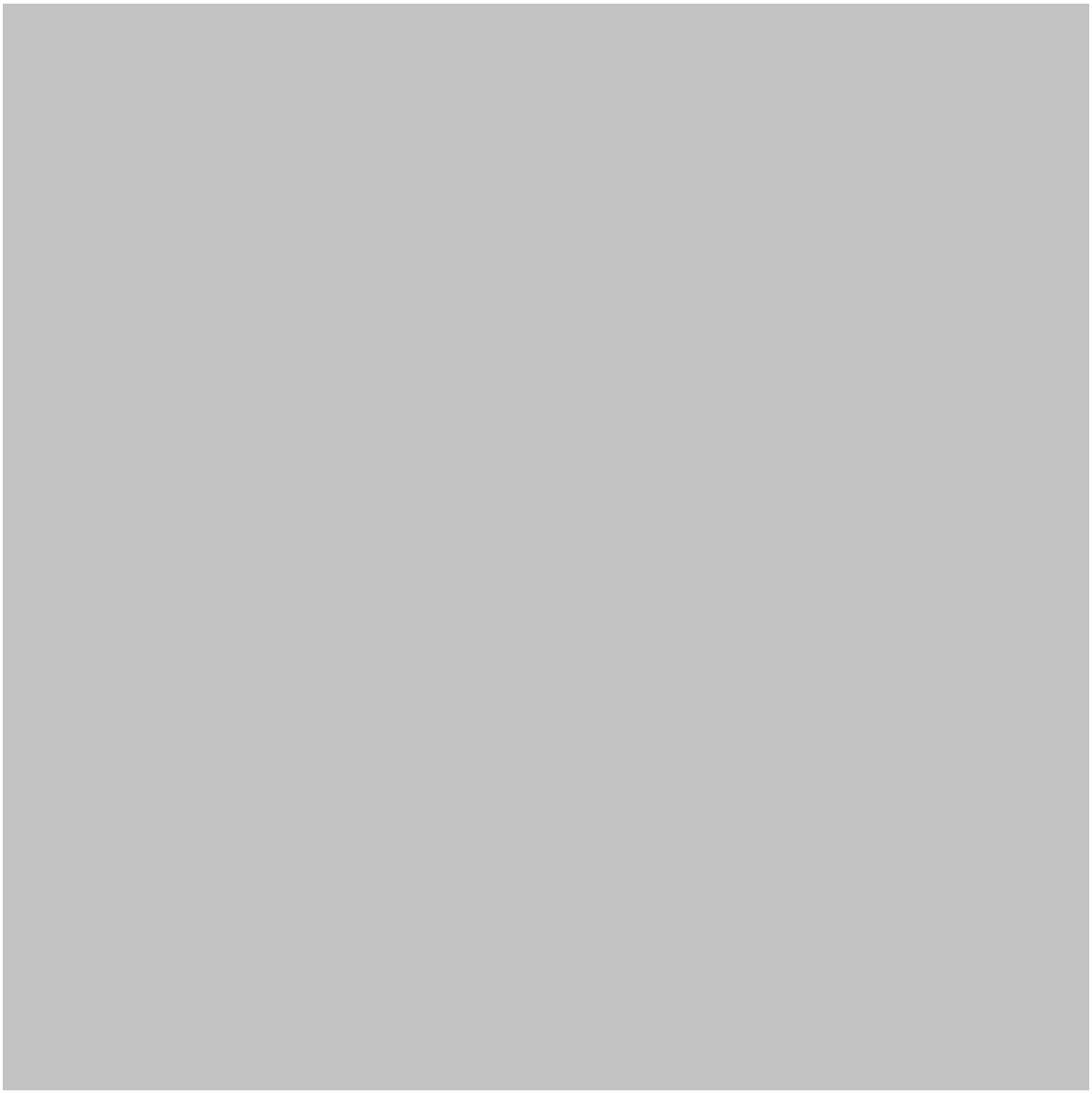
✅ Measurement precision improves.

✅ Instruments can be validated scientifically.

Step-by-Step Rasch Analysis in R 🛠️







Step 1 — Install Required Packages

Common packages include:

- eRm
- TAM
- psych
- mirt
- WrightMap

Example:

```
install.packages("eRm")
install.packages("psych")
install.packages("TAM")
install.packages("WrightMap")
```

Step 2 — Load Libraries

```
library(eRm)
library(psych)
library(TAM)
library(WrightMap)
```

Step 3 — Import Data

```
data <- read.csv("responses.csv")
```

Example structure:

Person	Q1	Q2	Q3	Q4
P1	1	0	1	1
P2	1	1	1	0
P3	0	0	1	0

Each row represents a respondent.

Each column represents one assessment item.

Step 4 — Fit the Rasch Model

```
model <- RM(data)
summary(model)
```

The model estimates:

- Item difficulty
- Standard errors
- Fit statistics

Step 5 — Evaluate Item Fit

```
itemfit(model)
```

Look for:

- Outfit Mean Square
- Infit Mean Square

Ideal values usually lie close to:

1.0

Acceptable range:

0.7–1.3

Step 6 — Estimate Person Parameters

```
person.parameter(model)
```

Outputs include:

- Ability estimates
- Reliability
- Standard errors

Step 7 — Plot Wright Map

```
plotPImap(model)
```

The Wright Map compares:

- Person ability
- Item difficulty

on the same measurement scale.

Step 8 — Interpret Results

Questions include:

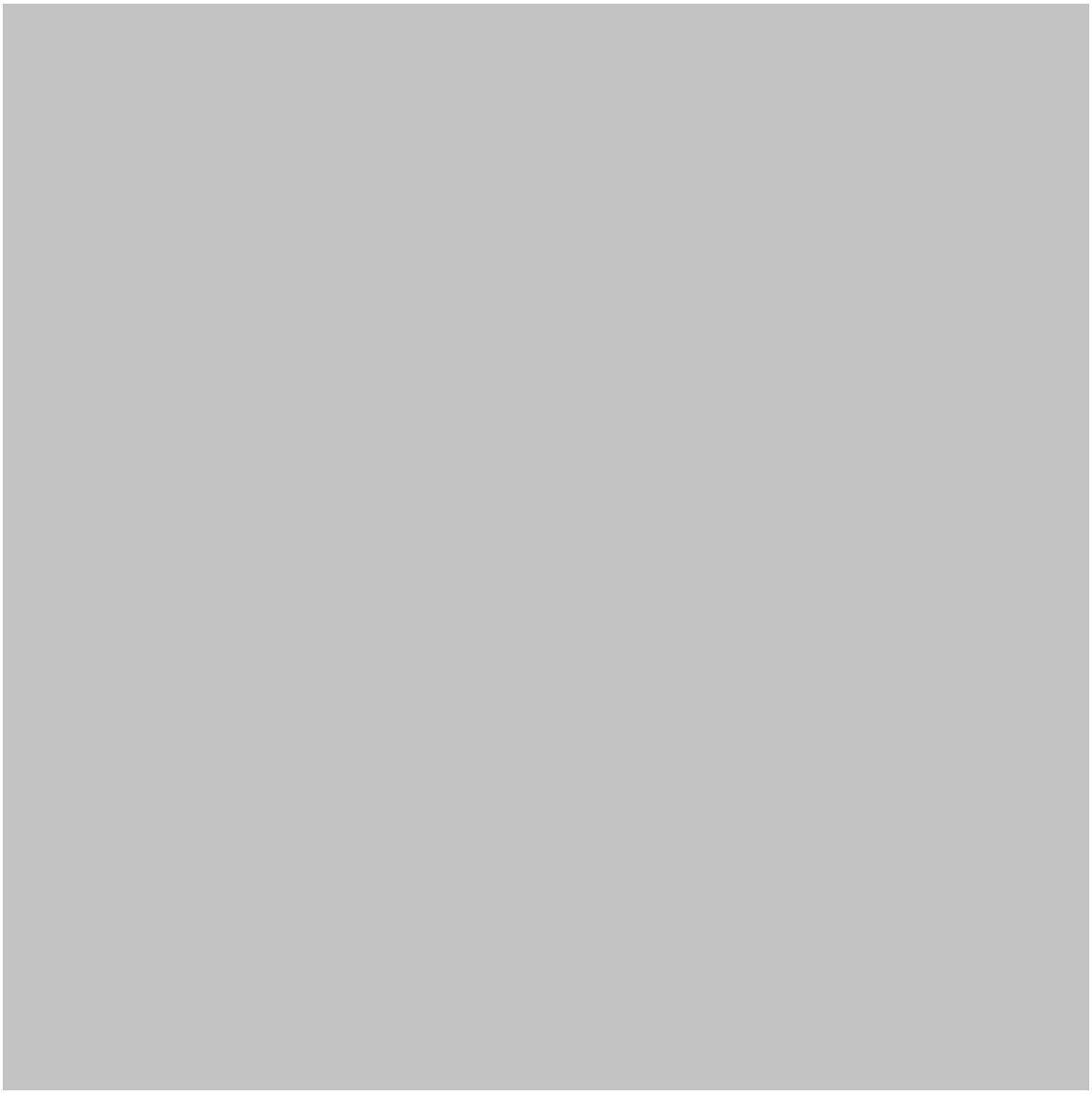
- Which items are easiest?

- Which items are hardest?
- Are items too easy?
- Are respondents well targeted?
- Does the test measure one construct?

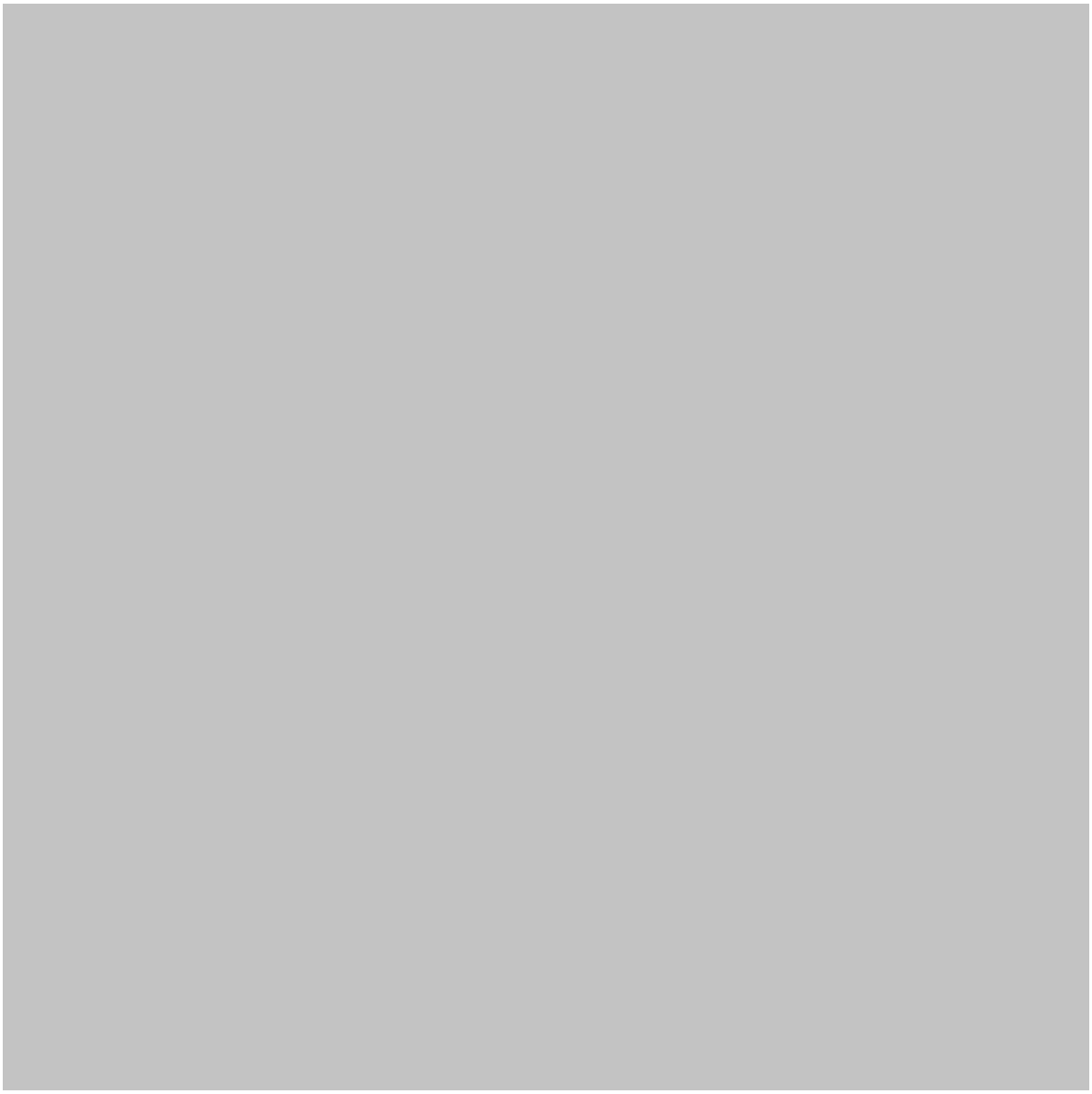
Comparison ⚖️

Feature	Classical Test Theory	Rasch Measurement Theory
Raw Scores	Yes	No
Item Difficulty	Limited	Estimated
Person Ability	Approximate	Estimated
Interval Scale	No	Yes
Missing Data	Problematic	More Robust
Instrument Validation	Limited	Strong
Measurement Precision	Moderate	High
International Comparisons	Difficult	Easier

Diagrams and Tables 📊







Typical Rasch Workflow

Stage	Purpose
Data Collection	Gather responses
Cleaning	Remove invalid records
Rasch Model	Estimate parameters
Fit Analysis	Check quality
Reliability	Evaluate consistency
Wright Map	Compare persons and items
Interpretation	Draw conclusions

Common Fit Statistics

Statistic	Good Value
Infit MNSQ	0.7–1.3
Outfit MNSQ	0.7–1.3
Person Reliability	>0.80
Item Reliability	>0.90
Separation Index	>2

Rasch Analysis Pipeline

```
Questionnaire
      |
      ▼
Data Collection
      |
      ▼
Cleaning
      |
      ▼
Model Fitting
      |
      ▼
Item Fit
      |
      ▼
Person Ability
      |
      ▼
Reliability
      |
      ▼
Validation
```

Examples 💡

Example 1 — Engineering Certification

A company evaluates 400 engineers using 50 technical questions.

Rasch identifies:

- 8 questions too easy
- 4 questions too difficult
- 38 performing well

The assessment becomes more balanced.

Example 2 — University Examination

A professor analyzes a programming exam.

The model reveals:

- Certain questions fail to distinguish high-performing students.
- Several items exhibit poor fit.

These questions are revised before the next semester.

Example 3 — Customer Satisfaction Survey

A manufacturing company measures customer satisfaction.

Rasch converts ordinal survey responses into interval measurements, enabling more accurate statistical comparisons.

Real-World Applications 🌍

Rasch Measurement Theory is widely used across numerous disciplines.

Engineering

- Competency assessments
- Technical certification
- Safety training evaluation
- Manufacturing quality control

Healthcare

- Patient-reported outcome measures
- Pain scales
- Rehabilitation assessments

Education

- National examinations
- University testing
- Online learning analytics

Human Resources

- Recruitment testing
- Leadership assessments
- Employee development

Psychology

- Personality scales
- Anxiety measurement
- Cognitive testing

Market Research

- Customer experience
- Brand loyalty
- Product satisfaction

Common Mistakes ❌

Many beginners make avoidable errors.

Using Small Samples

Very small datasets produce unstable estimates.

Ignoring Misfitting Items

Poorly fitting questions reduce measurement quality.

Treating Ordinal Scores as Interval Data

Raw Likert scores are ordinal.

Rasch converts them into interval measurements.

Overlooking Dimensionality

A Rasch model assumes one dominant latent construct.

Ignoring Reliability

Always evaluate:

- Person reliability
- Item reliability
- Separation indices

Challenges & Solutions 🔧

Challenge	Solution
Missing responses	Use Rasch estimation
Poor item wording	Rewrite questions
Low reliability	Add quality items
Misfit	Remove problematic items
Small sample	Increase respondents
Ceiling effects	Add harder items
Floor effects	Add easier items

Case Study 🏭

Engineering Competency Assessment

An aerospace company designs a technical competency examination for newly hired engineers.

Dataset

- 650 engineers
- 70 questions
- Binary responses

Initial Findings

Several questions show:

- Very low discrimination
- Poor wording
- Redundancy

Rasch Analysis

Results indicate:

- 12 misfitting items
- 7 items too easy
- 5 items too difficult

Improvements

The assessment is redesigned.

Outcomes include:

- Higher reliability
- Better targeting
- Improved fairness
- More accurate competency measurement

The organization now uses the revised assessment globally across multiple engineering departments.

Essential Tips ⭐

✔ Begin with clean data.

✔ Ensure adequate sample size.

📊 Examine Infit and Outfit statistics.

✔ Review Wright Maps carefully.

✔ Remove or revise poor items.

📊 Check unidimensionality.

✔ Report reliability indices.

✔ Document every modeling decision.

📊 Compare multiple Rasch models when appropriate.

✔ Validate findings using independent datasets whenever possible.

Frequently Asked Questions ❓

1. What is Rasch Measurement Theory?

It is a probabilistic model that estimates person ability and item difficulty on the same measurement scale.

2. Why use R for Rasch analysis?

R is free, powerful, highly extensible, and offers specialized packages for psychometric modeling and visualization.

3. Is Rasch better than Classical Test Theory?

For many measurement and instrument-validation tasks, Rasch provides stronger measurement properties because it estimates latent traits and item difficulty independently.

4. Which R package is most popular?

The **eRm** package is one of the most widely used for basic Rasch analysis, while **TAM** and **mirt** support more advanced modeling.

5. What sample size is recommended?

Although analyses can be performed with smaller datasets, many practitioners aim for **200 or more respondents** to obtain more stable item estimates, with larger samples preferred for complex instruments.

6. Can Rasch analyze Likert-scale questionnaires?

Yes. Extensions of the Rasch model, such as rating scale and partial credit models, are specifically designed for polytomous (Likert-type) responses.

7. What is a Wright Map?

A Wright Map displays respondent abilities and item difficulties on the same logit scale, helping researchers evaluate how well test items target the population.

Conclusion 🎓

Rasch Measurement Theory has transformed modern measurement by providing a rigorous, objective framework for evaluating tests, surveys, and assessments. Rather than relying solely on raw scores, it models the relationship between respondent ability and item difficulty, producing interval-level estimates that support more reliable decision-making.

When combined with the flexibility of **R**, researchers and engineers gain access to a powerful, cost-effective environment for conducting psychometric analyses, validating instruments, detecting problematic items, estimating reliability, and generating informative visualizations such as Wright Maps. Whether you are developing engineering competency exams, educational assessments, healthcare questionnaires, or customer satisfaction surveys, mastering Rasch analysis in R enables you to build fairer, more accurate, and scientifically defensible measurement systems.

By following a structured workflow—preparing high-quality data, fitting appropriate Rasch models, evaluating fit statistics, interpreting person and item parameters, and refining instruments based on evidence—you can significantly improve the quality and credibility of your research. As data-driven decision-making continues to expand across engineering, education, healthcare, and industry, Rasch Measurement Theory remains one of the most valuable tools for transforming raw responses into meaningful, reliable, and actionable measurements.

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